

Indian Statistical Institute
M.Tech. (CS), Final of Second Semester Examination, 2023-24
Database Management Systems

Full Marks: 50

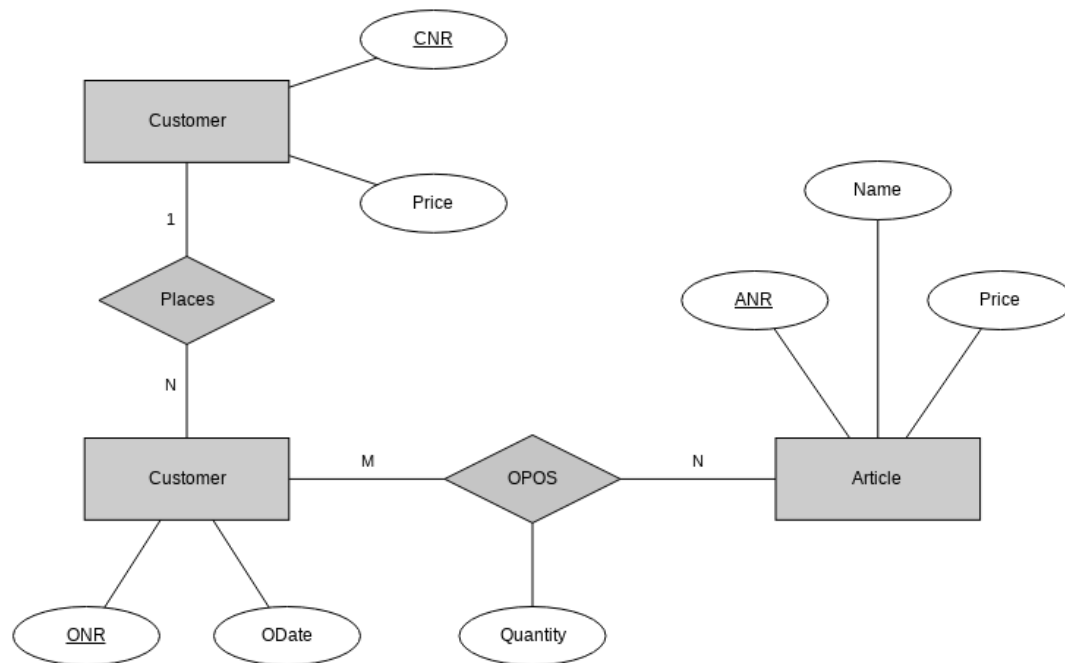
Date: 30-04-2024

Time: 2.5 Hours

Answer any *five* of the following questions

$5 \times 10 = 50$

1. (a) How can data type checking, format checking, and range checking help in the data validation process?
(b) Recall that semijoin ($\bowtie$) is similar to natural join with the only exception that attributes in the first relation are returned in the result. Conversely, antijoin ($\bar{\bowtie}$) returns all tuples in the first relation such that there are no tuples in the second relation with matching values for the shared attributes. Represent semijoin operation with antijoin and other relational algebra constructs. In other words, if R_1 and R_2 are two relations, then express $R_1 \bowtie R_2$ using $\bar{\bowtie}$ and other relational algebra operators.
(c) What are the advantages of relational algebra being (i) procedural and (ii) closed. 3+3+(2+2)
2. (a) Convert the following entity-relationship diagram into appropriate relational schema highlighting the primary keys, wherever applicable.



- (b) Justify the statement “Searching for a record within dense indices is faster than sparse indices, however, they require more space and impose higher maintenance overhead for

insertions and deletions”.

(c) Cite a useful application of the unclustered hash indexing strategy.

5+3+2

3. (a) Let there be a relational schema $R = \langle A, B, C, D, E, F \rangle$ that comprises only atomic attributes and holds the set of functional dependencies given as follows.

$$\{A \rightarrow BC, D \rightarrow E, E \rightarrow F\}.$$

What is the candidate key for R ? What is the highest normal form that R satisfies?

Justify your answer.

(b) Why is the functional dependency not preserved in Boyce-Codd normal form?

(c) How can you test whether the decomposition of a relation is lossless-join or not?

(1+3)+3+3

4. (a) Consider the pair of relations $R_1 = \langle A, B, C \rangle$ and $R_2 = \langle C, D, E \rangle$ having 1234 and 123 tuples, respectively. If the number of distinct entries of C in the tables R_1 and R_2 are 617 and 41, respectively, estimate the size of the following query.

$$\pi_C(R_1 \bowtie R_2)$$

(b) Prove or disprove the following equivalence relation of relational algebra.

$$\sigma_\theta(R_1 \cap R_2) \equiv \sigma_\theta(R_1) \cap \sigma_\theta(R_2)$$

6+4

5. (a) State and explain whether the given schedule, which abide by the two-phase locking protocol, will experience the following problems or not.

(i) deadlock

(ii) starvation

(iii) dirty read.

T_1	T_2	T_3	Timestamp
lock-X(B)			1
	lock-X(A)		2
	read(A)		3
	$A = A + A * 0.2$		4
	write(A)		5
	unlock(A)		6
		lock-X(A)	7
		read(A)	8
		$A = A - 100$	9
		write(A)	10
		unlock(A)	11
lock-X(A)			12
read(A)			13
$A = A + 10$			14
write(A)			15
downgrade(A)			16
		lock-S(B)	17
		lock-X(A)	18

- (b) Suppose the timestamp ordering protocol is applied for the multiversion scheme of concurrency control. The following is a sequence of versions stored for a particular data item, say A .

(10, 1, 2), (16, 3, 5), (14, 7, 6), (15, 8, 9)

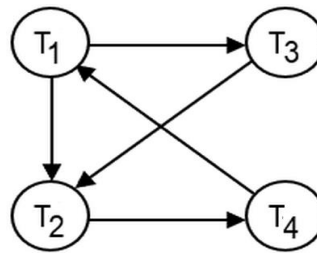
If a transaction T_1 has read the value of A as 10 and then 14 and a transaction T_2 has read the value of A as 16 only, which transaction has entered the system earlier? Justify your answer.

(2+2+2)+(2+2)

6. (a) Cite the following things with appropriate examples.

- (i) A conflict serializable schedule that is view serializable.
- (ii) A conflict serializable schedule that is not view serializable.
- (iii) A view serializable schedule that is not conflict serializable.

(b) The precedence graph of a schedule of 4 transactions is shown below. Justify whether it is conflict serializable or not. If it is conflict serializable, find out the serializability order of the schedule.



(2+2+2)+(2+2)

7. (a) Let there be a pair of tables having r_1 and r_2 number of rows and c_1 and c_2 number of columns, respectively. Suppose both the said tables have exactly one column as common and natural join is performed on them. If the results are stored in JSON format, then determine the minimum and maximum possible number of JSON documents that will be created? How many key-value pairs will each of these documents include?
- (b) Consider the following vector clock, which uses timestamps for concurrency control, for managing 4 nodes (numbered as 1, 2, 3 and 4).

2	6	8	9
16	12	14	15
1	3	4	5
10	7	13	11

Suppose an update happens on the node 3 at timestamp 17. Show and explain how the entire vector clock will get updated.

(2+2)+6